

Vinay Karande

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SUMMARY

I'm **Principal Software and Data Engineer** with almost **8 years** of experience leading the design and delivery of **scalable** backend, data, and **GenAI platforms** across **AdTech**, **EdTech**, and **Identity Systems**. Currently **leading cross-functional teams** and driving **end-to-end ownership** of complex systems—from architecture and design to implementation and production rollout. Hands-on in building **LLM-powered systems**, **LangChain Orchestration**, and **Milvus-based** vector search, while also **mentoring** engineers and influencing technical direction. Strong expertise in **Java**, **Spring**, **Kafka**, **Spark** and building realtime streaming, high-throughput, low-latency, fault-tolerant systems.

EXPERIENCE

Principal Software Engineer

Sep 2022 – Present

PubMatic (Pune, India)

- Leading the AI initiative to build an end-to-end **agentic AI system** using **LangGraph** and **Langfuse** for deal debugging and auto-healing, driving monetization through close collaboration with Product and clear requirement synthesis; designed **MCP-based integrations** with 20 data sources to analyze deal data and identify root causes of underperformance.
- Managing and mentoring a full scrum team of **9–11 engineers**, including direct reports, while owning delivery, technical direction, and execution.
- BURL**: Designed and implemented a real-time, high-throughput system processing **240M records/day** from **Kafka** using Confluent Parallel Consumers; deployed Dockerized services on **Kubernetes across 10 data centers**, achieving **3K QPS** with **sub-3-second SLA** per data center, and built comprehensive monitoring and alerting with **ELK** and Slack integrations.
- Architected distributed **Spring-based systems**, producing both high-level and low-level designs and achieving **3K QPS** through scalable design patterns.
- Built multiple end-to-end **Spark ETL pipelines**, loading data into **Snowflake** and exposing insights via Spring Boot-based UI applications.

Senior Software Engineer

Jun 2021 – Aug 2022

WhiteHat Jr. / BYJU'S Future School (Mumbai, India)

- Designed and built the complete **backend architecture** for **English Course features** from scratch and successfully launched them in the **beta phase**, while also leading the backend team for the “Show and Tell” feature through delivery.
- Led **backend development** for BYJU'S Future School, supporting **scalable APIs** for Coding, Math, English, and Arts courses across **Android and iOS platforms**.
- Implemented an **automated thumbnail generation framework** capable of extracting thumbnails from **videos, PDFs, and YouTube content**.
- Collaborated closely with **cross-functional stakeholders** to translate product requirements into **end-to-end technical solutions** and deliver them within timelines.

Software Engineer

Oct 2018 – May 2021

eQ Technologic Pvt. Ltd. (Pune, India)

- Designed and implemented an enterprise-grade **OAuth 2.0 (RFC 6749)** authorization framework and scalable **SSO authentication** systems, delivering Security-as-a-Service and Data Access Management solutions using SAML, Kerberos, SPNEGO, NTLM, certificate-based authentication, and LDAP.
- Enhanced platform security and reliability by contributing to enterprise-wide SSO, custom data access layers, and data migration engines, while proactively analyzing security risks and driving mitigation strategies for core products.

ACHIEVEMENTS

- 1st Prize – Hackathon (Deal Doctor, PubMatic)** – May 15, 2025
- Innovation Award** for building the Sherlock Framework end-to-end pipeline from scratch – June 05, 2025
- Innovation Award** for building the BURL solution (Spring Boot + Kafka parallel consumer) – March 03, 2025
- Biased Towards Action Award** for delivering multi-format solution in existing system – March 14, 2024
- Best Demo Award** for Kafka-based email queuing system design – April 06, 2023

TECHNICAL SKILLS

Languages & Frameworks: Java, Python, Scala, Spring Boot, FastAPI, LangChain, LangGraph

Data & DevOps: Kafka, Spark, Hadoop, MySQL, Redis, Snowflake, Milvus, Docker, Kubernetes, Jenkins, Grafana

KEY PROJECTS

- Agentic AI Deal Debugging:** Built GenAI system using LangGraph for automated deal analysis.
- Kafka System (BURL):** Designed parallel consumer based system processing 240M records/day.
- Middleware Backend:** End to End Spring Boot based Web Application to fetch data from Snowflake for Reporting.

EDUCATION

College of Engineering Pune

Pune, 2015 - 2018

B.Tech in Computer Engineering

Government Polytechnic Mumbai

Mumbai, 2012 - 2015

Diploma in Computer Engineering